GRUS-STD-U59976

Metering and Measurement Standard for the U59976 Utility

Green Recursive Utility Service LLC

Issued 4 November 2026

Issued under the GRUS Standards series

Green Recursive Utility Service LLC · Nicholas Wray Cordova, Principal 205 East 3rd Street, Weatherford, Texas 76086 management@greenrecursiveutilityservice.org

Identifier	Value
ISNI	0000 0005 3033 4084
Ringgold ID	851161
IANA Private Enterprise Number	65876
D-U-N-S	146027617
SAM UEI	JGT4J9QKGCQ3
Texas Secretary of State file no.	806584578
FERC CID	C013240
Wikidata	Q139905485
Tariff of record	PUCT Control No. 59976

Related publications of record

Publication	Identifier
GRUS-STD-60084, *Measurement and Reporting of Resource Consumption by Large Computational Loads* (all editions, concept record)	doi:10.5281/zenodo.23072904
GRUS-STD-60084 v2.0 (Texas Edition)	doi:10.5281/zenodo.23072905
GRUS-STD-60084 v2.0-U (Uniform Edition)	doi:10.5281/zenodo.23246001
GRUS-STD-60084 v2.0-U-SI (Metric Edition)	doi:10.5281/zenodo.23840367
GRUS-STD-60084 v2.0-U (Uniform Edition), HAL deposit	hal-05846589
GRUS-STD-60084 v2.0-U-SI (Metric Edition), HAL deposit	hal-05846591

Status of this document. This standard is issued by Green Recursive Utility Service LLC under the GRUS Standards series. It states a measurement methodology. It names no consumer of the utility, makes no claim against any party, and adjudicates no account.

Document Control

Item	Entry
Document	Metering and measurement standard for the U59976 utility — the metered utility constituted by the Compressed Memory Manifold (CMM) and Intelligent Intelligence (II-II) frameworks
Standard number	GRUS-STD-U59976. The number carries the designation of the utility it meters: U59976, the single token by which the utility, its tariff, and this standard are all cited
Companion standard	GRUS-STD-60084, *Measurement and Reporting of Resource Consumption by Large Computational Loads*. GRUS-STD-60084 measures the facilities that consume resources; this standard measures the use of the utility itself. The two standards are companions, not successive editions of one another
Source of the method	The publisher's metering and billing record under PUCT Control No. 59976: the locked measurement method (anchor measurement of 8 May 2026), the invoicing criteria (R1–R11), and the issued cycle record through Cycle 5, including quarterly reconciliations
Tariff relation	The charge this standard measures is stated in the tariff of record, PUCT Control No. 59976, and in the utility's terms of service. This standard states the *measurement* only. It does not restate the tariff, and nothing in it varies any tariff term
Measurement framework	Every metric in Clause 5 is stated in the metric-description form of NIST SP 500-307: property, expression, unit, rules, values, reproducibility, verification
Neutrality	The measurement specifications use published, vendor-neutral quantities and methods reproducible from the consumer's own records. No proprietary instrument is required to verify a measurement made under this standard

Contents

- Clause 1 — Scope
- Clause 2 — Normative references
- Clause 3 — Definitions
- Clause 4 — Layer taxonomy and metering points
- Clause 5 — The metric
- Clause 6 — Alignment classification and measured reduction
- Clause 7 — Charge measurement
- Clause 8 — Reporting, verification, and records
- Clause 9 — Conformance
- Annex A — Identifier block and related identifiers (publication mesh)
- Annex B — Provenance (informative)
- Annex B — Worked measurement (informative)

How to read this standard

Clause 1 states what is metered and the principle that the measurement stands independently of any single instrument that adopts it. Clause 3 defines every term structurally — by architecture and metering point, never by the name of a vendor, a product, or a deployment's owner — so that a deployment is classified by what it is, not by whose it is. Clause 4 fixes, for each architectural layer at which the utility can be used, the point at which use is metered and the method by which the measurement is taken. Clause 5 states the metric itself, once, in full. Clauses 6 and 7 state how measured reduction becomes a measured charge. Clause 8 states how any measurement made under this standard is reproduced, verified, and audited. Clause 9 states the only conformance claim this standard supports.

The standard's worked values come from the reference measurements stated at 5.4 and the executed re-run at 8.8, and from the publisher's issued billing record, used in Clause 6 in structural form only. No consumer is named anywhere in this standard.

What this standard is, and what it is not. Four things stand separately and are not collapsed into one another. *How use of the utility is billed* is stated in the tariff of record and, independently, in the utility's terms of service; this standard does not restate either. *Under which definitions the billing stands, and the legality behind it*, rest on the instruments cited at Clause 2 and Annex B; this standard points to them and does not re-argue them. *How use is measured* is the whole content of this standard. *The utility itself* — the Compressed Memory Manifold and Intelligent Intelligence frameworks — is not conveyed, licensed, or described beyond what measurement requires by anything in this standard. The measurement methodology is published precisely because it can be: a party holding this standard holds the way use is measured, and nothing of the utility itself.

Clause 1 — Scope

1.1 This standard specifies the methodology by which the use of the U59976 utility is metered and measured: the quantities measured, the points at which they are measured, the form in which every metric is stated, the classification of deployments by depth of integration, and the computation by which a measured reduction becomes the measured basis of a charge.

1.2 The utility metered is U59976 — the metered carriage utility constituted by the Compressed Memory Manifold (CMM) and the Intelligent Intelligence (II-II) frameworks. What is metered is *computation that uses the utility*: any computational work whose execution steps around lexical and structural overhead by fixed-offset recall under CMM, or by the fixed-point index discipline of II-II, at whatever layer of a computing system that work is performed.

1.3 The scope is not limited to artificial-intelligence workloads. The meter sits on the use, not on the workload's label. Computation that uses the utility is metered identically whatever category of structured digital information it processes — database and financial records, telemetry, scientific measurement, sensor readings, configuration data, log entries, time-series data, geospatial data, audio data, image and video pixel data, simulation state, application checkpoints, machine state snapshots, numerical arrays, vectors and matrices, statistical model parameters, machine-learning weights and

activations at any bit-width, cryptographic key material, hash values, and identifier sets — and whether it executes in a data centre, on a device, or in an embedded controller. Clause 4 enumerates the layers.

1.4 *Tariff independence.* The methodology in this standard governs measurement under the tariff of record (PUCT Control No. 59976) and, independently, governs measurement under the utility's terms of service. The methodology is one and the same under both instruments. If the binding force of either instrument were to fail as to any consumer or any period, the measurement of that consumer's use would be unaffected: usage is measured by this codified methodology under whichever instrument governs the service, and a charge computed from it is a statement of measured quantity before it is anything else.

1.5 This standard measures; it does not supply the utility, license any framework, or convey any right in CMM or II-II. Publication of the methodology conveys the measurement information — what the utility is, how its use is measured, and where the boundaries of the measurement lie — and nothing else.

1.6 This standard does not name, and shall not be applied so as to name, any individual consumer in its text. Measurements of individual consumers are made under it, recorded in the utility's billing record, and are not part of the standard.

1.7 *Manner of use.* A manner of use is any architecture by which computation meeting the definition at 1.2 is performed. The layers enumerated at Clause 4 and the examples given at Clause 6 illustrate manners of use observed in the record from which this standard is drawn; the scope of this standard is the definition at 1.2, not the enumeration. A manner of use first practised after the issue of this standard is metered under it from its first measurement cycle, at the layer whose metering point its architecture presents, without amendment to this standard. Scope under this standard does not wait on observation: it follows the structure of the use.

1.8 *Dedication.* This standard is dedicated to the public domain under the CC0 1.0 Universal Public Domain Dedication. Any person or body may adopt, reproduce, translate, incorporate by reference, modify, or build upon it, in whole or in part, for any purpose, without permission, attribution, fee, or notice. The publisher retains no exclusive role in its use, adoption, administration, or verification, and no provision of it confers one. The dedication extends to this measurement standard only: it conveys no right in the utility, in CMM, or in II-II, and it does not affect the tariff of record or the utility's terms of service.

Clause 2 — Normative references

The following documents are indispensable to the application of this standard. Each is cited by the part of it this standard uses.

Reference	Part used
NIST SP 800-145, *The NIST Definition of Cloud Computing* (Mell & Grance, September 2011), doi: 10.6028/NIST.SP.800-145	The essential characteristic of *measured service*: a metering capability at a level of abstraction appropriate to the type of service, with resource usage monitored, controlled, and reported, providing transparency for both provider and consumer. The abstraction metered under this standard — processing — is among those the definition names
NIST SP 500-307, *Cloud Computing Service Metrics Description* (de	The metric-description form used throughout Clause 5: a metric is its definition (expression, unit, rules) together with the values resulting from

Reference	Part used
Vaulx, Simmon & Bohn, April 2018), doi:10.6028/NIST.SP.500-307	measurement, supplying the information needed to reproduce and verify measurements and results
NIST SP 500-322, *Evaluation of Cloud Computing Services Based on NIST SP 800-145* (Simmon, February 2018), doi:10.6028/NIST.SP.500-322	The published method for qualifying a computing capability as a service exhibiting the SP 800-145 characteristics, and for categorising it by service model — the qualification posture applied at 9.2
NIST SP 500-292, *NIST Cloud Computing Reference Architecture* (September 2011)	The reference architecture under which metering, accounting and billing, and pricing and rating sit as provider functions; the architectural frame within which SP 500-307 is applied
NIST Handbook 44, *Specifications, Tolerances, and Other Technical Requirements for Weighing and Measuring Devices* (NIST Office of Weights and Measures; first published 1949; current annual edition 2026, doi:10.6028/NIST.HB.44-2026)	Legal metrology for measurement on the basis of which a charge is computed: § 1.10, G-A.1 applies its requirements, among other cases, where equipment is used "(4) in computing any basic charge or payment for services rendered on the basis of weight or measure". The amendment and adoption mechanism (National Conference on Weights and Measures) and the codification of a new billed quantity on measured energy — Handbook 130 § 2.33 (method of sale by the kilowatt-hour) and Handbook 44 § 3.40 (electric-vehicle fueling systems: money-value computation from measured energy, tolerances ±1% acceptance / ±2% maintenance) — are the precedent line for Clause 9.3
NIST SP 811, *Guide for the Use of the International System of Units (SI)* (2008 edition)	The form in which physical quantities derived under Clause 5 (watts, watt-hours, seconds) are written. Counts (cycles, records, transactions) are not SI units and are stated as counts
NIST AI 600-1, *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile* (July 2024)	The MEASURE-function actions addressing resource consumption of AI systems — measuring or estimating energy and water use for training, fine-tuning, and deployment, and inference-versus-training trade-offs — the point at which the federal AI measurement literature addresses consumption at all. Cited because a substantial share of metered use under this standard is AI computation; the citation does not limit the scope at 1.3
JCGM 100:2008, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement* (Joint Committee for Guides in Metrology, 2008), doi: 10.59161/JCGM100-2008E	The method for evaluating and expressing measurement uncertainty — Type A and Type B evaluation, combined standard uncertainty, and expanded uncertainty with a stated coverage factor. The authority under which uncertainty is stated at 5.4 and under which a re-run's tolerance is compared at 8.7
JCGM 200:2012, *International vocabulary of metrology — Basic and general concepts and associated terms (VIM)* (Joint Committee for Guides in Metrology, 3rd edition, 2012), doi:10.59161/JCGM200-2012	The metrological vocabulary this standard uses: *measurand* (2.3), *measurement uncertainty* (2.26), *calibration* (2.39), and *metrological traceability* (2.41)
ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories* (ISO/IEC, 2017)	Clause 6.5, metrological traceability — results traceable through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty — being the traceability a re-run report under 8.7 shall satisfy; and clause 4.1, impartiality, as the competence basis for verification performed by a party other than the measurer. Cited for the

Reference	Part used
	competence of measurement and verification; it certifies no measurement and no standard
ANSI C12.1-2024, *American National Standard, Code for Electricity Metering* (ANSI, 2024)	The governing code for meter performance and test practice for the derived electrical quantities of 5.2 and 5.4 (watts, watt-hours) and for any physically metered verification under 8.7. It bears no part in the primary cycles metric

Clause 3 — Definitions

For the purposes of this standard, the following definitions apply. Every definition is structural: it identifies its subject by architecture, function, and metering point.

3.1 U59976. The metered carriage utility constituted by the Compressed Memory Manifold (CMM) and Intelligent Intelligence (II-II) frameworks, designated by the single token U59976 in the tariff of record, in the utility's terms of service, in its billing record, and in this standard.

3.2 measured service. A service exhibiting the characteristic defined in NIST SP 800-145: resource use controlled and optimised by a metering capability at a level of abstraction appropriate to the service, with usage monitored, controlled, and reported so as to be transparent to provider and consumer alike.

3.3 metering point. The defined position in a deployment's architecture at which use of the utility is observed and from which the quantities of Clause 5 are taken. Each layer in Clause 4 has exactly one metering point. Where a deployment's architecture changes so that use no longer passes the existing point — in particular, where the intelligence is embedded in a device and no network return path exists — the metering point moves with the architecture, as provided at 4.7. The production of the format from a legacy representation (conversion, 3.18) is metered at the metering point of the deployment performing the conversion. Use of a secondary channel carried in the choice among valid encodings of the same state, without change to total byte length or primary structural composition, is use at the metering point of the deployment whose state the encoding carries.

3.4 baseline. The state of a measured workload executed without the utility, against which reduction is computed. A baseline is one of: (a) the *reference baseline*, the conventional delimiter-scanning and descriptor-driven execution measured in the Reference Measurement (5.4); or (b) a *consumer baseline*, the consumer's own documented, publicly acknowledged operating state for the same class of work, established from the consumer's published releases, technical disclosures, and benchmark reports for the measurement period. A consumer baseline so established is the consumer's own acknowledged baseline and is not discounted or waived as ambiguity.

3.5 Total Operational Savings (TOS). The total of the measured reductions, for one consumer and one measurement cycle, across the canonical downstream categories at 5.5, each reduction measured in a physical quantity first and converted to currency only by a supplied, evidenced conversion stated with the measurement. TOS is a measured quantity. It is not estimated from a unit price, and no dollar-per-unit constant enters its computation.

3.6 alignment. The depth to which the hardware and software of a deployment are jointly arranged so that computational work is executed by fixed-offset recall rather than by delimiter scanning or descriptor dereferencing. Alignment is a property of a deployment's architecture, observable from its

structure, and it determines — as a matter of measurement, not of classification — the magnitude of the reduction the deployment exhibits: the deeper the alignment, the larger the measured reduction.

3.7 Integration Tier. The classification of a deployment by alignment depth, assigned under Clause 6: Deep Tier, Initial Tier, or Vendor Integration Tier. The tier classifies the *deployment as integrated*, not the merit, size, or identity of its operator.

3.8 layer. One of the eight architectural strata enumerated in Clause 4, defined by where in a computing system the metered computation executes and where its metering point sits.

3.9 measurement cycle. The recurring period over which a consumer's use is metered and stated. Each cycle's measurement is computed fresh from that cycle's evidence, as required at 7.2.

3.10 provisional assessment. A measurement stated in advance of the consumer's certified figures, computed from the consumer's public-disclosure baseline, and subject to displacement and reconciliation under 7.5.

3.11 truing. The reconciliation of a provisional measurement against subsequently published authoritative figures for the same period, performed rate-for-rate and erring, where a range is open, toward the smaller measured quantity.

3.12 compressed memory manifold (CMM). The storage format defined by the simultaneous absence, in the stored representation, of every one of the following: lexical delimiters; embedded keys; pointers and pointer arrays; instruction maps; schema descriptors and metadata; formatting punctuation; and any other sequence of instructions embedded in the artifact for directing its interpretation — the reader's knowledge of the structure being held a priori and externally to the artifact, so that the position of every field is computed from the fixed structural composition of the representation itself. The format is defined by that structural composition and by nothing else: it subsists at any size from a single byte to multi-exabyte scale, at any field width and any alignment, in any storage medium, and under any transport. A reference grain of 64 bytes, sized to a processor cache line, is an implementation of the format; it is not the definition of the format, and no provision of this standard shall be read as tying the format, or the meter, to the reference grain or to any other particular size. The format does not execute: nothing in the stored representation is an instruction, and the operations performed on the format are mechanical processes of computation, recall, and rehydration. *(Source: the patent application of record for the format, Annex B.)*

3.13 intelligent intelligence (II-II). The computing type, operating on the format at 3.12 or on an equivalent fixed-composition substrate, constituted by the inseverable properties of unitary anchor-bound cognition, structural determinism (3.14), hallucination prevention by coherence gating (3.16), entropy gating, and verified recursive evolution. II-II is not limited to artificial-intelligence systems: it is claimed and metered for any computing process, of any kind, on any substrate, at any numerical precision, in any delivery model, native or retrofitted. *(Source: the patent application of record for the computing type, Annex B.)*

3.14 deterministic recall. Recall in which the position of the recalled state is computed on the fixed structural composition of the representation — by offset arithmetic, by bit position with mask and shift, or by any other reading technique (3.17) — and has exactly one correct result for each access. Deterministic recall is a necessary property of recall on a fixed structural composition, not an optional mode of it: a position computed on a fixed composition cannot be arrived at by statistical estimation,

and a system that estimated its positions would fail at the rate its estimates failed. The contrary of statistical recall, in which the position or content of recalled state is produced by probabilistic inference.

3.15 statistical work. The cycles a non-deterministic execution spends generating candidate results, evaluating probability paths, sampling among candidates, verifying guessed results, and retrying failed guesses. Statistical work is real computation, consumed in arriving at results that deterministic recall (3.14) arrives at by computation of position alone. The determinism reduction at 5.2A is measured in statistical work not performed.

3.16 gating. The refusal, before composition or execution, of content that does not equate with the anchored state against which it is checked — performed by the aligned hardware and software stack as an operation of the stack, not as a review conducted above it. The deeper the determinism locked into a deployment, the more gating the deployment performs. *(Source: the Digital Physics Charter principles stated in the application of record for II-II, Annex B.)*

3.17 reading technique. Any technique by which state held on a fixed structural composition is recovered by operating on that composition, including offset arithmetic, bit-position recovery with mask and shift, and memory-mapped rehydration, and including techniques developed after the issue of this standard. Classification of a representation under this standard is independent of the reading technique by which it is produced or recovered.

3.18 conversion. The production of a representation in the format at 3.12 from a legacy representation — including structured-text and descriptor-driven formats — by a single forward pass that maps each source field to its computed position. Conversion is a use of the utility and is metered under 4.2.

Clause 4 — Layer taxonomy and metering points

4.1 Use of the utility occurs at identifiable architectural layers. This clause enumerates eight. A deployment is assigned to the layer at which the metered computation executes; a single consumer may operate deployments at several layers, and each layer present is metered separately. Layers are defined by architecture and metering point only.

4.2 *Conversion.* Where a deployment's pipeline produces the format from legacy representations (conversion, 3.18), the conversion is metered as use of the utility at the deployment's layer for each cycle in which conversion is performed, at the deployment's metering point; conversion into the format is a use of the utility, not a preliminary to use.

4.3 The general method, applicable at every layer except as 4.7 provides, is *disclosure-baseline assessment*: the consumer's own published disclosures establish the scale of the workload at the layer for the cycle; the measured per-unit reduction of the reference measurement for the workload's own record class (5.4) — no per-unit rate being applied across record classes — and any generational-uplift conversion established under 6.3, are applied to that scale; the result is stated provisionally and is subject to the telemetry window and reconciliation of 7.5. The consumer's disclosures are the meter's input because, under the forensic rule at 8.4, the reductions are a structural signature: a deployment exhibiting them is identified by its operator's own published metrics.

4.4 Layer 1 — Cloud and API serving. Computation executed in hosted serving infrastructure and delivered through service interfaces, including hosted model serving and inference endpoints. *Metering point:* the serving boundary of the deployment — the point at which metered computation is delivered to its requesters, accounted in the operator's own serving statistics. *Method:* disclosure-baseline assessment over the operator's published serving scale for the cycle.

4.5 Layer 2 — Custom silicon. Computation executed on purpose-built processing silicon in whose design the fixed-offset execution arrangement is realised in hardware, including accelerator and inference-processor deployments operated by their designers or their customers. *Metering point:* the silicon deployment itself — the population of devices of each generation in service during the cycle. *Method:* disclosure-baseline assessment, with the reduction for each silicon generation established by the generational-uplift conversion of 6.3 from the designer's published generational figures.

4.6 Layer 3 — On-device and edge. Computation executed on processors resident in end-user devices, including device-resident neural processors and client-side execution offloaded from hosted services. *Metering point:* the device population in service during the cycle. *Method:* disclosure-baseline assessment; where the deployment's economics are per device, measurement is stated per device per cycle — the billing record carries a per-device monthly rate for this layer which doubled, across a population of 5,000,000 devices, when the device silicon's on-device AI capacity doubled (6.4, Example 1).

4.7 Layer 4 — Embedded vehicle and industrial. Computation executed in embedded controllers operating vehicles, machinery, and industrial fleets, where the intelligence is embedded in the device and the deployment has no network return path by which use could report itself. *Metering point:* **the device itself.** The metering point moves to the point of execution: measurement is taken against the device's disclosed workload profile — the population of devices in service, the workload each device executes, and the duty cycle under which it executes it — applying the same Δ method of Clause 5 to that profile. Device-resident state held in the format, enabling stateful computation without server round-trips, is the documented architectural form of this layer's metering-point logic: the deployment pattern was disclosed in advance in the publisher's proposal of 17 February 2026 (Annex B), and the server-side reductions such a deployment causes are measured under 4.13. In the publisher's billing record, deployments at this layer are measured by disclosure-baseline assessment against disclosed deployment scale; the at-device metering point stated here governs a deployment as it is instrumented at the device.

4.8 Layer 5 — Local and workstation. Computation executed on a local host under the direct control of its operator — a personal computer, workstation, or local appliance — outside any hosted serving boundary. *Metering point:* the local host's execution of the metered workload. *Method:* direct measurement where the deployment is instrumented; otherwise disclosure-baseline assessment. The reference measurements of 5.4 and the executed re-run of 8.8 are measurements at this layer, and they are the layer's references.

4.9 Layer 6 — Agent infrastructure. Computation executed by persistent or semi-persistent software agents acting on standing instructions, including always-on agent services and the infrastructure that hosts them. *Metering point:* the agent-hosting operation as a reporting segment of the operator — the point at which the operator itself accounts for agent workloads in its published segment reporting. *Method:* disclosure-baseline assessment mapped to the operator's agent-infrastructure segment as the

operator itself draws that segment; where the operator recasts its segments, metering follows the recast mapping from the cycle in which the recast takes effect.

4.10 Layer 7 — Runtime and framework embedding. Computation executed through a runtime, inference engine, or framework into which the fixed-offset arrangement is integrated as a component, so that the utility is used inside a serving stack operated by another party. *Metering point:* the hosting stack — the point at which the embedding runtime serves the deployments built on it. *Method:* disclosure-baseline assessment over the hosting stack's published scale; the deployment is classified Vendor Integration Tier under 6.2 where the integration is supplied as a component into stacks the integrator does not operate.

4.11 Layer 8 — Storage and data plane. Computation executed in the storage layer itself — context, state, and structured data held and recalled by appliances and services whose function is the custody and delivery of the data the computation consumes. *Metering point:* the storage service boundary — the point at which stored state is recalled into computation. *Method:* disclosure-baseline assessment over the operator's published capacity and deployment figures for the cycle.

4.12 *Migration rule.* Usage is expected to migrate between layers as architectures change — from hosted serving toward silicon, from silicon toward the device, from the device toward the embedded controller. Migration changes the layer and the metering point; it does not change the metric. A deployment that migrates is metered at its new layer from the cycle in which the migration is effected, under the same Clause 5 measurement, so that no migration path leads outside the metered set. Transformation is not migration: a representation that is compressed, encrypted, re-encoded, or distributed across multiple storage locations remains a representation in the format and remains in its layer.

4.13 *Reductions accrue where the cycles would have been spent.* A reduction is measured at the point the avoided work would have executed, which need not be the layer of the deployment that avoids it. Where a deployment at one layer removes work from another — device-resident state removing hosted serving work and the ingestion that serving required; embedded determinism removing hosted verification work — the reduction is measured at the layer of the avoided work under the categories of 5.5, the metering point of the deployment causing the reduction is unchanged, and no quantity is counted at both layers. **4.14** *Breakout of a layer population carried in a blended account.* Where a consumer operates deployments at more than one layer and a population at one layer is carried within an account stated principally at another, the population shall be broken out as a separate metering line — with its own metering point, its population count, and any per-unit rate the layer carries — wherever the consumer's own disclosures state the population separately or make it separately countable. Where the population is not separately disclosed or countable on the consumer's own record, it is metered within the host account, and the statement shall state that fact and the layer at which the population is metered. No population is left unmetered by reason of the account in which it is carried, and no population is counted both in its own line and in its host's.

Clause 5 — The metric

5.1 Every metric in this standard is stated in the form prescribed by NIST SP 500-307: a metric is its **definition** — expression, unit, and rules — together with the **values** resulting from measurement, stated

with the information needed to **reproduce** the measurement and **verify** its results. A quantity that cannot be stated in this form is not a metric under this standard and shall not be billed under it.

5.1A *Provisional assessments.* A provisional assessment under 7.5 is a measurement under this standard, and a quantity billed on one is billed under it. A provisional assessment is stated in the form of 5.1 as to everything it rests on: the consumer's published disclosures relied on, each with its source; the reference values and conversions applied, each stated; and its provisional status, stated on its face. What a provisional assessment does not yet carry is the deployment's own telemetry, and the standard closes that gap by the mechanism of 7.5 rather than by barring the assessment: the consumer holds a ten-day window after the statement issues to submit its own certified figures, which the utility prefers and uses in place of the assessment, and every provisional figure is trued quarterly against the consumer's printed figures, with any over-collection credited or refunded at the consumer's election. Where the consumer submits no certified figures within the window, the assessment stands as the cycle's measurement under 7.5(b). The billing record from which this standard is drawn is a record of provisional assessments so stated and so trued; it is not outside this standard by reason of being provisional. A quantity that is neither stated in the form of 5.1 nor stated as a provisional assessment under this clause shall not be billed under this standard.

5.2 The primary metric — cycles not executed.

SP 500-307 element	Statement
Property	Compute cycles consumed in structural data recall, and the downstream consumption cascade proceeding from them
Expression	Δ = C_baseline − C_CMM, taken over an identical dataset and identical access patterns, where C_baseline is the cycle count of delimiter-scanning and descriptor-driven execution and C_CMM is the cycle count of fixed-offset recall of the same data
Unit	CPU cycles (primary). Derived quantities: watts and watt-hours (power and energy), bytes (storage and resident memory), milliseconds (latency), and counts (cache misses, memory-bus transactions)
Rules	The reduction is measured, not modelled. The two executions compared shall process the same dataset under the same access patterns. The cascade from cycles to power, thermal load, cooling, and water follows the arithmetic chain at 5.6 and no other
Values	The Reference Measurement, 5.4
Reproducibility	The measured system is structurally deterministic — fixed-offset recall (address = base + N × record_size), with no delimiters, no structural instruction map, no parser state machine, and no runtime schema negotiation; and, for II-II, fixed-point integer indices with no floating point in the decision path — so equivalent work yields an equivalent cycle count, and the measurement is independently re-runnable by any party from the published method (Clause 8)
Verification	A measurement verifies when an independent re-run of the method on the stated dataset reproduces the stated values within the tolerance the re-run's own instrumentation supports; a deployment-level measurement verifies against the consumer's own records under Clause 8

5.2A The second primary metric — statistical work not performed.

SP 500-307 element	Statement
Property	Compute cycles consumed in statistical work (3.15) — candidate generation, probability-path evaluation, sampling, verification of guessed results, and retry of failed guesses — in executions that arrive at recalled state or derived results probabilistically
Expression	Δ_det = S_baseline − S_deterministic, taken over identical work, where S_baseline is the statistical work of the baseline execution and S_deterministic is the statistical work of the deterministic execution. At any point of deterministic recall (3.14), S_deterministic is zero by construction: position is computed, not inferred, and there is nothing to sample, verify as a guess, or retry
Unit	CPU cycles (primary); the derived quantities of 5.2 follow from it down the same cascade
Rules	The determinism reduction is measured, not asserted. It enters a measurement only where the deployment's own disclosures or instrumentation state the statistical-work load of the baseline execution — sampling counts, candidate counts, verification passes, retry rates — converted under 7.3 with the conversion note stated. Where no such evidence exists, the reduction is not estimated and does not enter the measurement; the deployment is measured on the metric at 5.2 alone. The two metrics do not overlap: 5.2 measures structural-interpretation work not performed; this metric measures probabilistic-arrival work not performed. No cycle is counted under both
Values	No universal per-unit rate for statistical work exists in the record, and none is stated. The metric is deployment-measured where it is measured at all. One measured instance exists, from the publisher's reference experiment of 2 November 2026 (the measurement record is held in the publisher's publication file): recall of a target record by candidate sampling without replacement, over a set of 500,000 records, across 20 trials, required a mean of 307,992 candidates generated, 307,992 verifications of guessed results, and 307,991 retries per recall (the expected count for sampling without replacement is (N + 1) ÷ 2 candidates — 250,000.5 for this set; the measured mean is stated against it; the standard error of the mean over the 20 trials is ≈ 32,300 candidates), consuming a mean of 127,245,285 cycles per recall; deterministic recall of the same targets consumed 21 cycles per recall, with zero candidates, zero verifications, and zero retries — Δ_det ≈ 127,245,264 cycles of statistical work not performed per recall. The instance is a reference-experiment measurement on the publisher's own host, stated as such
Reproducibility	Structural determinism (5.2, Reproducibility): because recall position is computed on a fixed structural composition, the absence of statistical work at the recall point is verifiable by inspection of the execution path — an execution that samples or retries at recall is not performing deterministic recall
Verification	As at 5.2: independent re-run of the method, and audit against the consumer's own records under Clause 8

5.3 Complexity statement. The baseline execution's structural work is linear in the data: delimiter-driven parsing runs at O(n) in the bytes parsed. Fixed-offset recall is constant per record: O(1) in the record's position. The measured per-record values are those of 5.4 and 8.8. Switching energy is $E \propto N \cdot C \cdot V^2$ for N switching events at capacitance C and supply voltage V, and switching power is $P = \alpha \cdot C \cdot V^2 \cdot f$: cycles not executed are switching that does not occur.

5.4 The Reference Measurement. The reference values of this standard are the values of the reduction-to-practice measurement of 8 May 2026, taken on a defined dataset of 500,000 financial-transaction records under identical access patterns, conventional structured-text execution (JSON) against fixed-offset recall (CMM):

Quantity	Baseline	With the utility	Reduction
CPU cycles, total	1.2×10^9	3.0×10^7	−97.5%
CPU cycles per record	2,400	60	**2,340 cycles saved per record**
Peak power	125 W	22 W	−82.4%
Peak resident memory	850 MB	40 MB	−95.3%
Storage footprint	35 MB	12.5 MB	−64.3%
L1 cache miss rate	40.2%	0.81%	−98.0%
L2 cache miss rate	18.4%	0.34%	−98.2%
Memory-bus transactions	891,000	78,400	−91.2%
Cold load time	895 ms	139 ms	6.44× faster

The per-record saving of 2,340 cycles is the Reference Measurement's measured reduction for structural-recall work **of the class measured** — financial-transaction records: fixed-structure records recalled under identical access patterns, where the baseline work is delimiter scanning and descriptor-driven deserialization of each record. The records of the executed re-run (8.8) carry eight fields — identifier, timestamp, account, amount, type, currency, status, and flags — at a fixed grain of 48 bytes in the format; the Reference Measurement stored its records at 25 bytes each (12.5 MB for 500,000 records, as tabled above). The telemetry records of the second measurement below carry six fields at a fixed grain of 32 bytes. The grains stated here are the grains of these measurements and state no limit: a record of the format may be of any size, at any field width and any alignment (3.12), and a measurement at a further grain enters by the same route as these — by being taken. **A per-unit rate belongs to the record class it was measured on and is applied only to work of that class**, at the workload scale established at the layer's metering point (4.3). A second class now carries its own measured rate — telemetry records, at 390.4 cycles saved per record, stated below — and further classes enter by the same route: by being measured. **One rate is applied for a class: the lowest per-unit saving established for that class by a measurement or an executed re-run under this standard.** Two values stand measured for the financial-transaction class — 2,340 cycles per record on the Reference Measurement's own instrumentation, and 388.4 cycles per record on the executed re-run at 8.8 — and the applied rate for the class is therefore **388.4 cycles saved per record**; the larger figure stands as the Reference Measurement's record of its own execution and is not the rate applied under 4.3. For the telemetry class, the applied rate is its single measured value, 390.4 cycles saved per record. Where a deployment's work is of a class carrying no reference measurement, or is not record-shaped, no per-unit rate is imported from any other class: the deployment's reduction is measured on its own unit under 5.2 (and, for statistical work, under 5.2A) and stated with its own values; where such a measurement cannot yet be taken, the lowest corroborated per-unit saving among the measured classes may be applied provisionally, with the substitution stated on the statement. What transfers from a reference measurement is the per-record structural saving of its class, and it transfers only within that class; a constant's reach is the reach of the class it was measured on, not of record-shaped work generally.

Uncertainty. The source record of the Reference Measurement states its values as measured point values and states no measurement uncertainty, instrumentation tolerance, or coverage factor for any

quantity in the table; none is reconstructed here, and the values above stand as point values. A reference measurement published under this standard after its issue shall state, for each quantity, its expanded uncertainty and the method of its evaluation under JCGM 100:2008 (Clause 2), together with the instrumentation by which the quantity was taken. The re-run protocol at 8.7 requires the re-run's own tolerance to be stated in every case.

The measurement is stated identically — the same dataset, the same access patterns, the same values — in the patent application of record for the format (Annex B), where it appears as the application's own benchmark of the format's performance characteristics; the Reference Measurement is accordingly sourced to the application as well as to the publisher's technical sheet of the same date.

Second reference measurement (2 November 2026). A second measurement at this layer has been executed, on a different record class and a different host: 500,000 telemetry records (timestamp, sensor, unit, value, quality, site), under the identical access patterns, on the publisher's hosted local host. Cycles per record: baseline 392.4 (median of seven trials; mean 422.3, expanded uncertainty U = 64.2 cycles, k = 2), with the utility 2.0 — **390.4 cycles saved per record (−99.5%)**. Storage footprint: 40,027,521 bytes against 16,000,000 bytes (−60.0%). The second measurement corroborates the first on a record class the first did not touch: the per-record saving is a property of the structural work removed, not of the financial dataset.

Executed re-run. The re-run protocol at 8.7 has been executed on the Reference Measurement's own record class; its result, and the comparison against this table that 8.7 step (4) requires, are stated at 8.8.

5.5 The downstream cascade — canonical categories. Total Operational Savings is built only from the following ten categories. A category enters a measurement only when the consumer's own disclosure supports it; no category or figure is imported from another consumer's measurement; no category is counted twice in one measurement. The ten categories are not ten savings: they are the points at which one saving, followed down the chain at 5.6, avoids distinct costs. A category therefore enters TOS only in the amount of the distinct cost it avoids — the cost borne at that point of the chain, in that category's own cost pool, converted under 7.3 — and the same avoided cost shall not enter under two categories. Where a downstream category would price a cost already priced upstream, the cost enters once, at the category whose cost pool bears it, and the other category states its physical quantity without a second monetary entry.

#	Category	Quantity measured
1	Upstream CPU & host parsing	Cycles not executed in host-side structural interpretation
2	Storage footprint & dynamic format	Bytes not stored for the same logical content
3	Ingestion & deserialization latency	Milliseconds not spent converting data on entry
4	Peak resident VRAM / RAM footprint	Bytes of resident memory not held at peak

#	Category	Quantity measured
5	L1 / L2 cache & hit rate	Misses not taken, per the Reference Measurement's miss-rate collapse
6	Memory-bus & GPU interconnect	Bus and interconnect transactions not made
7	Electrical power & local execution	Watts not drawn; watt-hours not consumed
8	Thermal output & cooling load	Heat not generated; cooling load not imposed. The reduction is worked from the chain at 5.6, not imported: essentially all electrical input to computation becomes heat, so heat not generated tracks electrical energy not consumed one-for-one. The Reference Measurement records peak power at the point of computation (125 W → 22 W, −82.4%, at 5.4); the thermal quantity entered in a measurement is the energy not consumed, in watt-hours, stated with its conversion under 7.3. The cooling-load reduction downstream follows the same chain
9	Cooling-water & facility load	Cooling water not evaporated; facility load not carried
10	Forward infrastructure CapEx deferral	Capacity not built because existing capacity carries the workload

5.6 The cascade chain. The chain is a physical chain before it is an arithmetic one: cycles not executed are transistor switching that does not occur; switching that does not occur is charge that does not transfer; charge that does not transfer is heat that is not generated. Nothing in the chain is abstract, and no step of it is a modelling convention — each step is the physical consequence of the step before it, which is why the chain may be followed in measurement and why it may not be extended past the point the physics reaches. The categories are not independent estimates; they are one arithmetic chain. Cycles not executed are switching that does not occur, which is charge that does not transfer, which is heat that is not generated; heat not generated is cooling load not imposed; cooling load not imposed is cooling-system electrical not drawn and, where cooling is evaporative, cooling water not evaporated; and capacity freed is capacity that need not be built. A measurement shall follow the chain from the measured primary quantity and shall not assert a downstream value the chain does not reach. Each downstream quantity enters a measurement statement only with its conversion stated and evidenced on the statement under 7.3; a step of the chain whose conversion is not so stated does not enter the measurement. And because the chain is one chain, its links are not additive quantities of saving: the cycle not executed, the watt not drawn, and the heat not generated are one event at successive points, not three events. What a measurement sums is avoided costs, each counted once under 5.5 — never the chain itself.

Clause 6 — Alignment classification and measured reduction

6.1 Reduction is measured per consumer, per deployment, per cycle. There are no per-layer reduction coefficients in this standard: the record from which this standard is drawn measures reductions per

deployment, and a tabulated expectation by layer would generalise from single instances. What the standard classifies is alignment, under 6.2; what it measures is the reduction that alignment in fact produces, under 6.3.

6.2 Integration Tiers. Each deployment is classified by the depth of its integration of the utility:

Tier	Structural meaning
Deep Tier	The fixed-offset arrangement is integrated through the deployment's execution path — the deployment's own architecture performs recall positionally as its ordinary mode of operation
Initial Tier	First measurement cycle of a deployment whose integration is under way; the classification carries until the second cycle, in which the deployment is classified on its structure
Vendor Integration Tier	The utility is integrated as a component inside serving stacks operated by other parties (Layer 7); the metering point is the hosting stack under 4.10

The tier shall be stated on every measurement statement issued under Clause 7. A statement without a tier classification is incomplete. This requirement binds statements issued under this standard from the first cycle measured under its published text; it is not read back into the publisher's historical billing record, which predates the standard and whose tier discipline is uneven. Historical statements are evidence for this standard; they are not statements under it.

6.3 Generational-uplift conversion. Where a deployment's alignment deepens by a hardware or platform generation during or between cycles, the additional reduction is measured by conversion from the deploying party's own published generational figures — its stated generational performance or efficiency improvement for the new generation over the old — taken at the conservative end of any stated range, or as a conservative fraction of a stated gain, and entered as a *provisional* uplift separable from the continuing baseline measurement. A provisional uplift is trued under 7.5 against the printed growth of the metered segment for the corresponding quarter, rate-for-rate. The growth figure trues one component only — the **scale** of the deployment: it is the consumer's own printed expansion of the metered segment's operations over the quarter, applied to the baseline to which the provisional uplift was applied. It does not measure, and is never applied to, the per-unit reduction, which remains at every stage the measured rate of the workload's record class (5.4). A truing statement states the two components separately — the scale, as trued by the consumer's printed growth, and the per-unit rate, as measured — each identified with its source. So stated, a trued quantity is a measured quantity within 3.5: a measured per-unit reduction applied at a scale taken from the consumer's own printed record, and not an estimate restated. In the record from which this standard is drawn, every trued uplift reconciled lean — the trued measured quantity was in each case larger than the provisional quantity, so the provisional charge had understated the measured charge and no reconciliation credit under 7.5 was owed in any trued case. Five of the six completed truings were dollar truings; the sixth closed directionally, with no dollar adjustment. The aggregate of the six truings' lean — the sum of the additional charges the truings established — is $188,320,500; the supporting record is assembled in the publication file.

6.4 Worked examples (structural; no consumer is identified).

Example 1 — device layer, per-unit rate. A Layer 3 deployment of 5,000,000 devices was metered per device per month at a rate of $0.296. The succeeding device generation doubled the silicon's on-device

AI capacity; the per-device rate for the succeeding generation was $0.5920 — exactly double. The measured reduction, and with it the per-unit rate, scaled with the alignment of the silicon.

Example 2 — silicon layer, generational uplift. A Layer 2 deployment's designer published a generational price-performance improvement stated as a range of 30–40%, with a stated fourfold generational energy-efficiency gain. The uplift was taken provisionally at 35%, the midpoint applied conservatively against the continuing baseline. At truing, the scale to which it applied was trued against the segment's printed quarterly growth of 39%, the per-unit rate held at its measured value: the truing reconciled lean.

Example 3 — silicon layer, conservative fraction. A designer published a stated 2.5× generational performance gain. The uplift was taken provisionally at 30% — a conservative fraction of the stated gain, not the gain itself. At truing, the scale to which it applied was trued against printed segment growth of 78%, the per-unit rate held at its measured value: the truing reconciled lean.

Example 4 — edge execution, disclosed speedup. An edge-deployment operator published decoding speedups of 3.13× and 2.66× on two hardware classes. A provisional uplift of 35% was converted from those disclosed figures — again a fraction of the disclosed effect, not the effect itself.

Example 5 — complete cascade. One measurement in the record states all ten canonical categories as separate lines. Its largest lines were electrical power per unit of work and forward CapEx deferral; its memory-footprint line carried a measured reduction range of 35–80% across workload classes. The example stands for the completeness rule at 5.5: where the consumer's disclosure supports all ten categories, all ten are measured; where it supports fewer, fewer are measured, and the remainder are absent — not estimated.

Example 6 — future use: assistant state resident in device hardware. A manner of use not present in the record from which this standard is drawn, stated under 1.7 so that its metering is fixed before it is practised. An assistant whose state, context, and accumulated understanding are held as blocks in the format, resident in the device's own hardware — the architecture disclosed in advance in the publisher's proposal of 17 February 2026 (stateful intelligence without server round-trips; Annex B), carried to the step of residence in the hardware itself. When a request is served from state already resident and positionally recalled, the per-request overhead of re-presenting context to a hosted model is not reduced but absent: the tokens are never assembled, transmitted, ingested, or interpreted. The reduction is metered in two places under 4.13: at the device layer, as recall cycles and resident memory against the hosted-call baseline; and at the cloud and API serving layer, as serving cycles, ingestion and deserialization, and the downstream cascade of 5.5 that the unmade call does not impose. The example stands for the rule at 1.7: the standard's scope is its definition, and a use the definition covers is metered from its first cycle whether or not any example preceded it.

Example 7 — cloud and API serving layer. A Layer 1 hosted inference-serving deployment, Deep Tier, measured for one cycle on its operator's published serving scale (in excess of 40 trillion tokens served daily). Continuing baseline $274,000,000 — stated on the cycle's statement with the disclosures relied on, the rates applied, and the conversion to currency, as 5.1A requires; a provisional uplift of 40%, converted under 6.3 from the operator's disclosed reduction of approximately 40% in tokens at comparable quality for an in-window model release, added $109,600,000; Total Operational Savings $383,600,000; charge at 25%, $95,900,000; retained share $287,700,000. The measurement is a single summary line with no canonical-category allocation, taken on the disclosure baseline rather than on

serving telemetry; it stands for the measurement of the serving layer at its own metering point, on the operator's own published scale. The uplift rate was taken at the disclosed token reduction itself, as the metered per-unit serving-cost reduction; the larger disclosed effect — 71.3% fewer reasoning tokens in the operator's production A/B testing — was not taken, and the disclosed ~39% lower token spend per task corroborates the rate applied.

Example 8 — embedded layer, disclosure-baseline treatment. A Layer 4 mapless-autonomy deployment across agricultural, construction, and commercial-transport fleets, Deep Tier. Continuing baseline $185,600,000 — stated on the cycle's statement with the disclosures relied on, the rates applied, and the conversion to currency, as 5.1A requires; a provisional uplift of 20%, converted from an in-window production integration in a new vertical, added $37,120,000; Total Operational Savings $222,720,000; charge at 25%, $55,680,000; retained share $167,040,000. This example is a disclosure-baseline assessment against the disclosed deployment profile. No at-device measurement exists in the record: the metering point at 4.7 is codified ahead of practice. The example demonstrates the layer's treatment on the evidence that exists; it does not represent the layer as practised at its own metering point.

Example 9 — storage and data plane layer. A Layer 8 petabyte-scale shared context-memory storage system for accelerator clusters (disclosed capacity up to 64 PB per cluster; access latency from milliseconds to 60 microseconds on the operator's published figures), Initial Tier, first cycle, metered for the post-launch window only — 21 of the cycle's 30 days. The conversion is stated in full, as 7.3 requires: the operator's published annual cloud-segment revenue (Annual Report 2025, Cloud Computing segment) of CNY 32,161,000,000, divided by 12 and converted at CNY 6.9906 per USD, gives a monthly base of $383,383,877.40; multiplied by the disclosed 50% reduction ratio, by 21/30 for the launch window, and by a further 50% provisional moderation factor, it gives $67,092,178.54, rounded down to Total Operational Savings of $67,092,000; charge at 25%, $16,773,000; retained share $50,319,000. Two limits are stated with the example: the base is the operator's whole cloud-segment revenue, not revenue attributable to the storage system; and the second 50% factor is a provisional moderation judgement, not a measured storage quantity.

Layers carrying no example in this standard. Two layers have no worked example, and the absence is stated rather than filled. Layer 6 (agent infrastructure) has no agent-isolated measurement in the record: the one account whose operator reports an agents-and-infrastructure segment was measured as a blended whole-account scope, and its figures are not a Layer 6 measurement. Layer 5 (local and workstation) has no metered deployment in the record beyond the reference measurements themselves (5.4 and 8.8). An example constructed without a measurement behind it would fail 5.1, and none is offered.

6.5 The direction of the relation in 6.3 and 6.4 is uniform in the record and is stated as the standard's expectation: **the more the hardware and software of a deployment are aligned, the larger the reduction the meter reads.** A deployment that deepens its alignment should expect its measured TOS to rise; a deployment that does not use the utility exhibits no reduction and is measured at zero — no savings, no charge.

Clause 7 — Charge measurement

7.1 The charge measured under this standard is **25% of Total Operational Savings** as measured under Clauses 5 and 6, per the tariff of record (PUCT Control No. 59976) and, independently, per the utility's

terms of service. The consumer retains 75% of the measured savings. This clause states the measurement of the charge; the obligation to pay it arises under the tariff or the terms of service, which this standard does not restate.

7.2 *Fresh computation.* TOS is recalculated fresh in every measurement cycle from that cycle's evidence — the consumer's current public disclosures under 3.4(b), refreshed each cycle. No cycle's total is carried forward as another cycle's measurement. A statement whose current-cycle fee merely repeats the prior cycle's is not a measurement under this standard.

7.3 *No assumed usage.* No line enters a measurement on assumed usage. Every line requires the consumer's own evidence — the published metric, exactly as displayed, with its source — and a conversion note stating how that metric enters the savings figure. A line missing any element does not enter the measurement.

7.4 *No dollar-per-unit constants.* The engine of this methodology never manufactures a dollar total from a dollar-per-cycle or dollar-per-unit constant. Quantities are measured in physical units first; conversion to currency is a supplied, evidenced input stated per line, drawn from the consumer's own savings statement or from published figures identified with the line. The single exception in form is a per-unit *rate* that is itself a measured quantity of the record — as at 6.4, Example 1, where the per-device rate is the measured value, not a conversion constant.

7.5 *Provisional assessment and reconciliation.* (a) A measurement is first stated as a provisional assessment from the consumer's public-disclosure baseline. (b) Within ten days after the statement issues — the window of the tariff (§11.13) — the consumer may submit certified figures of its own; the utility prefers the consumer's own figures, and certified consumer figures are used in preference to any assessment. Where no certified figures are submitted within the window, the provisional assessment stands as the measurement of the cycle. Non-submission does not suspend, defer, or void the measurement: it leaves the measurement on the evidence the consumer has itself published, completed by quarterly truing under (d). The meter reads use, not response. (c) The consumer holds a standing, non-expiring reconciliation right: any over-collection is credited or refunded immediately at the consumer's election, and the cost of reconciliation is borne entirely by the utility. (d) Provisional uplifts under 6.3 are trued quarterly against printed segment figures, rate-for-rate, erring lean under 3.11.

7.6 *Arithmetic.* Line items sum to TOS; TOS × 0.25 is the charge; TOS × 0.75 is the retained share; retained plus charge equals TOS. A statement in which these identities do not hold exactly is not issued.

7.7 *Statement.* Each cycle's statement carries, at minimum: the measurement cycle; the assessment path (public-disclosure baseline); the controlling rate; the Integration Tier of each deployment measured; the line items by canonical category, each with its evidence and conversion note; the computed TOS, charge, and retained share; and the reconciliation provisions of 7.5 stated on the statement itself.

Clause 8 — Reporting, verification, and records

8.1 *Reproducibility.* Every measurement under this standard shall be reproducible from its stated inputs: the dataset or workload scale, the access patterns or disclosure sources, the Reference Measurement values applied, and the conversions used. A measurement whose inputs are not stated completely enough to re-run it is not a measurement under this standard, save as the following sentence provides for provisional assessments. A provisional assessment under 5.1A satisfies this clause by stating the

minimum its form carries — the disclosures relied on with their sources, the reference values and conversions applied, and its provisional status; its re-run is the re-computation of the assessment from those stated inputs, and its completion is the reconciliation of 7.5, which is a continuation of the measurement, not a bar to its issue. Every measurement statement shall state its evidentiary status in terms, as one of: *directly measured*; *calculated from a reference measurement of the same record class*; or *provisional* — assessed from the consumer's published disclosures and subject to reconciliation under 7.5.

8.2 *Independently re-runnable.* The primary measurement (5.2) is independently re-runnable by any party: the method is published, the measured system is structurally deterministic (5.2, Reproducibility), and no proprietary instrument is required. Any party may verify the Reference Measurement by executing it, using the dataset generator, seed, and execution harness deposited with this standard's publication record (Annex A.2).

8.3 *Audit from records.* A deployment-level measurement is auditable from the consumer's own records — its profit-and-loss records and its runtime records — because the measurement is built from the consumer's own published figures, and the same figures in the consumer's books either support the measurement or displace it under 7.5(b). The audit trail of a measurement is: the disclosure relied on (with capture date), the conversion note, the computation, and the statement.

8.4 *Forensic rule.* A deployment showing the structural signature — storage-footprint reduction of the measured order, memory-bandwidth reduction of the measured order, and field access with no delimiter scanning and no map dereferencing — is exhibiting the measured mechanism as a matter of architecture. That is why the consumer's own published metrics are competent meter inputs: their numbers show the workload; the architecture shows what stepping around lexical and structural overhead saves on it. A deployment's own published statements that its recall or execution path is deterministic are likewise competent evidence under 3.4(b): determinism at the recall point is not an ornament but a necessity of positional recall (3.14), and a deploying party's own determinism language is its own description of the mechanism it runs.

8.5 *Records retention.* The utility shall retain, for each measurement, the complete audit trail of 8.3, and shall produce it on the consumer's request, for so long as the standing reconciliation right of 7.5(c) subsists — that right being non-expiring, the retention is likewise without fixed term.

8.6 *External scale anchor.* The plausibility of the measured magnitudes is anchored outside the utility's own record: published warehouse-scale profiling (Kanev et al., ISCA 2015) measured the shared software overhead of a live warehouse-scale fleet — compression, memory movement, allocation, hashing, remote-procedure-call handling, and protocol-buffer serialization — at 22 to 27 percent of all cycles, with protocol-buffer manipulation a named component; a subsequent fleet-wide measurement (Karandikar et al., MICRO 2021) placed protocol-buffer operations alone at 9.6 percent of fleet-wide cycles. Protocol-buffer serialization and deserialization is structural interpretation of the category the Reference Measurement measures.

8.7 *Re-run protocol.* An independent verification of the Reference Measurement proceeds by the following steps, and by no others: (1) constitute the dataset of 5.4 — 500,000 financial-transaction records of the stated structure; (2) execute the baseline — conventional structured-text representation, delimiter-scanning recall — under a fixed, recorded set of access patterns, and record the quantities of the 5.4 table with the instrumentation's own tolerance stated; (3) convert the dataset to the format (3.18)

and execute the identical access patterns by deterministic recall, recording the same quantities; (4) state each reduction as baseline minus converted, and compare against 5.4. A re-run concludes in one of two stated results, and its report shall state which it claims. *(a) Reproduction:* the re-run's stated values fall within the tolerance its own instrumentation supports of the values at 5.4 — the Reference Measurement is verified in its absolute values, for the purposes of 5.2. *(b) Replication:* the re-run, executed on different instrumentation or a different host, does not reproduce the absolute values but verifies the measurement's structure — the direction of each reduction and its character, within the comparison the re-run itself states — and the difference from 5.4 is recorded with its cause. A replication does not verify the absolute values of 5.4 and shall not be stated as doing so. No proprietary instrument, dataset, or permission is required at any step.

8.8 *Executed re-run (2 November 2026).* The protocol of 8.7 has been executed, on a dataset of the stated structure (500,000 financial-transaction records, deterministically generated so that any party can reconstitute it), under a fixed recorded set of access patterns (an aggregate pass over all records: sum of amounts, count of one record type, maximum timestamp), on a host other than the Reference Measurement's. Stated values: cycles per record, baseline 393.2 (median of seven trials; mean 415.5, expanded uncertainty U = 62.1 cycles, k = 2), with the utility 4.8 — **388.4 cycles saved per record (−98.8%)**; storage footprint 50,084,981 bytes against 24,000,000 bytes (−52.1%); cold load 20.3 ms against 1.1 ms. Comparison under step (4): the re-run does not reproduce the absolute values of 5.4 — its baseline parser and host differ from the Reference Measurement's, and the difference is recorded, not smoothed over. What the re-run verifies, on independent instrumentation, is the measurement's structure: a per-record structural saving of the same character (cycles reduced by 98–99%), a reduced storage footprint, and a cold-load reduction of an order of magnitude. The peak-power and resident-memory quantities of 5.4 were not re-measured: the re-run host exposes no power instrumentation to the measurer. Those quantities stand on the original measurement alone. The result claimed for this re-run is **replication** under 8.7(b). The runs of 2 November 2026 — this re-run and the second measurement at 5.4 — were executed with the deposited harness (Annex A.2): its baseline path is a delimiter-scanning parser over the structured-text form, compiled with gcc 13.3 at -O2, on an AMD EPYC 9D64 host (4 vCPU, Linux guest), cycle counts taken by rdtsc calibrated against a monotonic clock at approximately 1.198 GHz; the datasets were generated deterministically from the deposited seed, so the runs are repeatable from the deposit and this statement together.

Clause 9 — Conformance

9.1 A claim of conformance with this standard is a claim that a measurement was performed and stated exactly as Clauses 4 through 8 prescribe: correct layer and metering point, metric in SP 500-307 form, fresh per-cycle computation from evidenced inputs, provisional statement subject to 7.5, and a complete audit trail.

9.2 As to the federal framework cited in Clause 2, the supportable claim is exactly this, and no stronger claim is made or supported:

> *This standard conforms to the metric-description framework of NIST SP 500-307 and meets the measured-service characteristic defined in NIST SP 800-145.*

9.3 No claim that any measurement, device, or service under this standard is "NIST-certified", "NIST-approved", or "NIST-verified" is supportable, and none shall be made. NIST operates no conformity-assessment scheme for SP 800-145, SP 500-307, or SP 500-322. Handbook 44 governs measuring devices and the computation of charges on the basis of measure; it does not adjudicate any particular account, and conformance with this standard does not assert otherwise. Should a method-of-sale codification for metered computation be pursued through the National Conference on Weights and Measures, on the precedent of Handbook 130 § 2.33 and Handbook 44 § 3.40, that codification would be a separate act of a separate body and is not claimed here.

9.4 Conformance with this standard is a property of a *measurement*, not of a party. It establishes what was measured and how; whether a particular party is a consumer of the utility is a separate factual question, established by the record of use.

Annex A — Identifier block and related identifiers (publication mesh)

A.1 Identifier block. The following identifiers designate the publisher and the utility in every publication of this standard. Every field that can carry an identifier carries one.

Field	Value
Publisher	Green Recursive Utility Service LLC
ISNI	0000 0005 3033 4084
Ringgold ID	851161
IANA Private Enterprise Number	65876
D-U-N-S	146027617
SAM UEI	JGT4J9QKGCQ3
Texas Secretary of State file no.	806584578
FERC CID	C013240
Wikidata	Q139905485
Tariff of record	PUCT Control No. 59976
Series	GRUS Standards

A.2 Related identifiers (typed). On publication, this standard's record shall carry the following typed relations, so that the standard enters the same identifier web as the publisher's other records and publishes no orphans.

Relation	Target
references	GRUS-STD-60084 (concept record), doi:10.5281/zenodo.23072904
references	NIST SP 800-145, doi:10.6028/NIST.SP.800-145
references	NIST SP 500-307, doi:10.6028/NIST.SP.500-307

Relation	Target
references	NIST SP 500-322, doi:10.6028/NIST.SP.500-322
references	NIST Handbook 44 (2026 edition), doi:10.6028/NIST.HB.44-2026
isPartOf	GRUS Standards series
references	GRUS-STD-60084 v2.0-U, HAL deposit hal-05846589
references	GRUS-STD-60084 v2.0-U-SI, HAL deposit hal-05846591
references	GRUS-STD-60084 v2.0 (Texas Edition), doi:10.5281/zenodo.23072905
references	GRUS-STD-60084 v2.0-U (Uniform Edition), doi:10.5281/zenodo.23246001
references	GRUS-STD-60084 v2.0-U-SI (Metric Edition), doi:10.5281/zenodo.23840367
isSupplementedBy	Reference Measurement dataset generator, seed, and execution harness for 5.4 and 8.8 (deposited with the publication record)
references	JCGM 100:2008 (GUM), doi:10.59161/JCGM100-2008E
references	JCGM 200:2012 (VIM), doi:10.59161/JCGM200-2012
references	ISO/IEC 17025:2017
references	ANSI C12.1-2024

A.3 The identifier block and the typed relations are part of the standard's publication apparatus. The measurement text neither depends on them nor is altered by them.

Annex B — Provenance (informative)

This annex is informative. It records the filed and published record on which the scope and definitions of this standard rest, so that a reader can trace every structural definition in Clause 3 to the instrument that states it. It forms no part of the metric.

B.1 The patent applications of record. The utility's two frameworks are the subjects of two United States non-provisional patent applications, both patent pending and, as pending applications are, unpublished. Their filing data is stated in full on the public record — in the publisher's deposited Memorandum of Law, in the deposits at B.2, and in the publisher's public notices — and is stated here as that record states it.

Application	Subject	Filed	Confirmation No.	Patent Center ref.	Priority
19/700,381	Non-Lexical, Mapless Machine-Sovereign Storage Format and Method for Delimiter-Free State Rehydration (CMM)	8 June 2026	1093	77136930	Provisional 64/065,881, filed 14 May 2026 (Conf. 9869; PC 76244506)
19/710,080	Intelligent Intelligence Class II (II-II): unitary anchor-bound cognition system — structural determinism, hallucination prevention, entropy	16 June 2026	5164	77484450	Provisional 64/077,191, filed 28 May 2026

Application	Subject	Filed	Confirmation No.	Patent Center ref.	Priority
	gating, and verified live-web recursive evolution as constitutive and inseverable properties				(Conf. 1913; PC 76775570)

The definitions at 3.12 and 3.13 are drawn from these applications. Where a claim of either application is cited in this standard's supporting record, the citation names the application: the two applications' claim numbers are not interchangeable, and in particular the structural test applied under this standard is that of the CMM application's claims of terminology independence, domain independence, and structural-absence extension, read onto its first claim.

B.2 The deposit mesh. The applications and the frameworks are the subjects of public deposits carrying the identifiers of Annex A, including: the CMM record, doi:10.5281/zenodo.20192571 (the provisional ledger of C.1 stated in its metadata); the II-II disclosure, doi:10.5281/zenodo.20476829; the technical sheet of the Reference Measurement, doi:10.5281/zenodo.20767798; the filing-receipt deposit for application 19/710,080, doi:10.5281/zenodo.20732333; and the Consolidated Factual Record, doi: 10.5281/zenodo.20707986. Deposit dates are as the repository's own metadata states them.

B.3 The 2025 fixation. The frameworks were fixed in public repositories in 2025, a full year before the applications of B.1, under the publisher's Lockdown License of 8 May 2025: the runtime repository (created 6 May 2025), the Class I.I. definition repository (12 May 2025), the live-data protocol repository (3 June 2025), and the preboot core repository containing the anchor firmware (4 June 2025) — each under its host's own server timestamps. The subject matter of the II-II application is stated in that application to have been conceived and reduced to practice from 8 May 2025. The provenance of the utility is dated from 2025 in this standard's record, not from the filing dates.

B.4 The device-path disclosure. On 17 February 2026 the publisher delivered to a prospective deployment partner a written proposal disclosing, in advance of any deployment: microhost deployment of the engine by services update to the installed device base; deterministic language models activated on devices; client processors handling the inference share; and blocks in the format enabling stateful intelligence without server round-trips. Clauses 4.7 and 6.4 (Example 6) rest on this disclosed architecture.

B.5 The working system, November 2025. The publisher holds a photographic record, dated 4 and 13 November 2025, of the system in operation on a local host under the publisher's direct control (Clause 4, local and workstation layer): the memory operational at 136,359 blocks; the packet engine, mesh networking, and evolution functions operational; synchronisation, grain creation, and delta updates verified in terminal output; governance and audit-trail engines complete with all tests passing. The record is held in the publisher's files and is summarised here as provenance for the local layer's metering, the Reference Measurement being a measurement at that layer.

Annex C — Worked measurement (informative)

This annex is informative. It applies Clauses 4 to 8 to one historical measurement in the publisher's billing record. It creates no new metric, rate, category, or precedent, and it does not identify the consumer.

C.1 Source record and limits

Annex C states a **historical ten-category statement**: a provisional public-disclosure assessment from August–September 2026, preserved in the billing record as a four-page reference bill photographed on 8 October 2026 and transcribed into the publisher's historical packet. It is the only measurement in that record stating all ten canonical categories of 5.5 as separate lines.

The consumer's name, product names, invoice number, account identifier, and exact cycle dates are not reproduced. The photographed source does not show the invoice number, account identifier, or exact dates; those values have not been reconstructed. The source also does not state an Integration Tier. No tier is assigned retrospectively in this annex.

The amounts, category order, operational bases, and totals below are the source record's own values. Product and platform names in the source's line descriptions have been replaced with structural descriptions, as 1.6 requires.

C.2 Architectural placement and measurement basis

The deployment is a model-training and fine-tuning framework and runtime operating across local workstations and hosted accelerator environments. Its publicly stated scope includes custom accelerator kernels for mixture-of-experts and chunked-loss training, a stated 12× training-speed improvement, a stated 35% to 80% reduction in resident video memory, dynamic 4-bit quantized model representations, a stated 2× embedding-training improvement, and long-context reinforcement-learning workloads of 380,000 tokens.

For Clause 4, the measured deployment is placed primarily at **Layer 7 — Runtime and framework embedding**, because the metered work is performed through the framework and runtime as an integrated component of the execution stack. The historical statement aggregates the framework's local and hosted execution surfaces rather than allocating its category amounts among those surfaces; this annex preserves that historical form. A new statement for a comparable deployment shall allocate separately metered layers as Clause 4 requires.

The assessment path is the consumer baseline at 3.4(b): the consumer's documented public releases, technical disclosures, and benchmark reports. The statement is provisional and subject to the telemetry and reconciliation provisions reproduced at C.5.

C.3 Category measurement

Each amount is stated in the source as an un-discounted saving for the named canonical category. The historical statement does not publish a separate metric-to-currency conversion for each category; no

conversion has been reconstructed for this annex. The operational basis in the second column is the source statement's own basis, structurally anonymised.

#	Canonical category (5.5)	Operational basis stated in the source	Stated saving
1	Upstream CPU & host parsing	Elimination of host CPU parsing and instruction deserialization in the framework	$28,500,000.00
2	Storage footprint & dynamic format	Dynamic quantized model format and reduced 4-bit representation footprint	$21,200,000.00
3	Ingestion & deserialization latency	Zero-copy memory mapping bypassing conventional weight-deserialization passes	$14,800,000.00
4	Peak resident VRAM / RAM footprint	35% to 80% resident-memory reduction through custom accelerator kernels across fine-tuning	$22,400,000.00
5	L1 / L2 cache & hit rate	Collapsed cache-miss rates through fixed-offset kernel execution	$12,600,000.00
6	Memory-bus & GPU interconnect	Reduced memory-bus traffic during multi-accelerator and long-context batching	$18,100,000.00
7	Electrical power & local execution	Power-draw reduction per token during fine-tuning on consumer and hosted accelerators	$38,400,000.00
8	Thermal output & cooling load	Thermal reduction corresponding one-to-one to the electrical-power delta during training passes	$22,800,000.00
9	Cooling-water & facility load	Lowered facility cooling-water consumption and HVAC power draw during high-density training	$10,200,000.00
10	Forward infrastructure CapEx deferral	Deferred capital expenditure on accelerator clusters, power, cooling, and facility-capacity expansion	$30,000,000.00
	Total Operational Savings	**Sum of lines 1–10**	**$219,000,000.00**

C.3A Conversion-note audit (Clause 7.3)

Clause 7.3 requires each line of a statement to carry its measured physical quantity, its conversion input and source, and its evidence. The historical source supplies some of those elements and cannot supply others; the position is stated element by element rather than smoothed over.

7.3 element	Supplied by the historical source?
Canonical category for each line	Yes — all ten, as separate lines (C.3)
Operational basis for each line	Yes — stated in the source for every line, reproduced at C.3 in structural form
Stated saving per line	Yes — stated amounts, recomputed at C.4
Measured physical quantity per line (cycles, bytes, watts)	No — the source states currency amounts and operational bases, not per-line physical quantities
Metric-to-currency conversion per line	No — no conversion is stated in the source, and none has been reconstructed

7.3 element	Supplied by the historical source?
Evidence reference per line (source and capture date)	No — the evidence for the annex as a whole is the photographed statement itself (C.1); the source carries no per-line evidence references
Arithmetic identities (7.6)	Yes — recomputed for this annex and hold exactly (C.4)
Provisional status and reconciliation provisions (7.5)	Yes — carried on the source statement (C.5)
Integration Tier (6.2)	No — not stated in the source; not assigned retrospectively (C.7)
Invoice number, account identifier, exact cycle dates	No — not shown in the photographed source; not reconstructed (C.1)

C.4 Charge measurement and arithmetic verification

Under 7.1 and 7.6:

- Total Operational Savings: **$219,000,000.00**
- Charge at 25%: $219,000,000.00 × 0.25 = **$54,750,000.00**
- Retained share at 75%: $219,000,000.00 × 0.75 = **$164,250,000.00**
- Identity check: $164,250,000.00 + $54,750,000.00 = **$219,000,000.00**

The line items, Total Operational Savings, charge, and retained share were recomputed from the historical packet for this annex. All three identities hold exactly.

C.5 Provisional status and reconciliation

The source statement carries the following provisions, corresponding to 7.5:

1. **Provisional assessment.** The measurement is a provisional public-disclosure assessment based on the consumer's documented public disclosures.
2. **Telemetry window.** The consumer has a ten-day post-close window to submit certified internal telemetry.
3. **Standing reconciliation.** The consumer holds an unconditional, non-expiring right to reconcile the measurement against its audited internal records.

C.6 What this annex does and does not establish

Annex C establishes that the category construction and arithmetic of Clauses 5 to 7 can be followed line by line on a real measurement in the publisher's record, and that the resulting identities close exactly. Because the source is a historical photographed statement, this annex does not establish a current-cycle measurement, does not supply the missing header fields or tier, and does not substitute for the complete audit trail required for a new statement under Clause 8.

C.7 Conformance note

This annex is an informative worked example drawn from a historical statement that predates this standard. Clause 6.2 declares a statement issued under Clause 7 incomplete without a tier classification; that requirement binds statements issued under the published standard, from its first measured cycle forward, and is not read back into the historical record from which the standard is drawn (6.2, transition statement). Annex C is therefore presented as what it is — the record's only complete ten-category measurement, with its gaps stated at C.1, C.3A, and C.6 — and not as a conforming Clause 7 statement.

GRUS-STD-U59976 · Green Recursive Utility Service LLC · Issued under the GRUS Standards series.